Concept Note: TransitAI

AI-Based Bus Demand Prediction and Route Planning System

1. Why — Problem Statement

Public transportation faces changing passenger demand throughout the day. Fixed bus allocation can cause overcrowding during peak hours and underutilization during low-demand periods. Manual demand estimation also makes efficient planning difficult. Therefore, an intelligent, data-driven system is needed to support better bus allocation and transportation planning.

2. What — Proposed Solution

TransitAI is a machine learning-based web application that predicts passenger demand and recommends the number of buses required based on estimated demand and bus capacity.

Key features include:

- Demand Prediction: Estimates passenger demand using selected input parameters.
- Smart Bus Allocation: Recommends the number of buses based on passenger capacity.
- Forecast and Dashboard: Visualizes demand patterns and prediction results.
- CSV Export: Allows users to download available data and results.
- Interactive Route Map: Displays bus stops and route sequences using Delhi GTFS data.

The system is developed using Python, Flask, machine learning libraries, and web-based mapping tools.

3. Whom — Target Users

TransitAI is intended for:

- Public Transport Authorities: To support transportation planning.
- Bus Operators: To improve fleet utilization.
- Fleet Managers: To estimate bus requirements.
- Urban Planners: To support smarter mobility planning.
- Researchers and Students: To explore AI-based transportation solutions.

Expected Outcome

TransitAI aims to demonstrate how machine learning can support passenger demand estimation, capacity-based bus allocation, and transit route visualization through a single web application. The current prototype uses synthetic demand data and a separate Delhi GTFS dataset; validation with real passenger data is a future step.